

Project Report: Yash Parking Valet Management System

1. Project Overview

The **Yash Parking Valet Management System** is a company-specific application designed to manage and organize vehicle parking inside the campus.

Most existing systems only record who enters the premises. This project goes a step further by managing **where each vehicle should park**, making parking structured and efficient.

2. Why This Project Is Different

Existing Approach:

- Security records vehicle entry manually or through access cards
- No tracking of parking slot allocation
- No real-time view of parking availability

Proposed System:

- Automatically assigns a specific parking slot to each vehicle
- Shows real-time availability of parking slots
- Helps security manage parking efficiently during peak hours

This system **does not replace entry systems** it complements them by managing parking operations.

3. Problem Statement

Currently, there is no digital system to manage parking slots inside the company premises. This results in:

- Time wasted searching for parking
- Confusion during peak hours
- Security staff not knowing actual parking availability
- Inefficient use of parking space

4. Project Objectives

- Assign a specific parking slot to every vehicle
- Display live parking availability on a dashboard
- Simplify parking management for the security team
- Reduce parking-related confusion and delays

5. System Workflow

1. **Vehicle Entry:** Security enters vehicle details into the application
2. **Slot Allocation:** The system automatically assigns an available slot
3. **Live Dashboard:** Parking availability updates in real time
4. **Vehicle Exit:** On exit, the slot is freed and made available again

6. Technology Stack

- **Python:** Core logic and parking calculations
- **Streamlit:** Web-based dashboard for the application
- **Lists & Dictionaries:** Temporary in-memory data storage

7. Future Scope

This project is designed to be scalable and can be enhanced further:

- **Database Integration:** Store parking data permanently
- **Reports & Analytics:** Identify peak parking hours and usage patterns
- **AI/ML Integration:**
 - Predict days or time slots when parking may become full
 - Analyze parking demand trends
- **Image Recognition:**
 - Automatic number plate recognition using cameras

8. Conclusion

The **Yash Parking Valet Management System** addresses a real operational problem within the company. It provides a structured, efficient, and scalable approach to parking management. The system starts with a simple implementation and offers strong potential for future enhancements using advanced technologies.